

Axial Force in a BOSEM Coil–Magnet Actuator

A Bulk-Current / Dipole-Density Derivation

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Abstract

This note derives, from Biot–Savart’s law, the exact axial-force integral between a coil and a coaxial permanent magnet. The coil is modeled as a bulk azimuthal current distribution; the magnet is modeled as a continuum of axial magnetic dipoles. Combining the two yields a five-dimensional integral with no elliptic-integral shortcuts taken, suitable for direct numerical evaluation by Gauss–Legendre quadrature.

1 Setup

The coil (inner radius R , outer radius R_a , length L) and the magnet (radius a , height h) share the z -axis. The coil is centered at $z = 0$; the magnet is centered at the scan position z_c (Fig. 1).

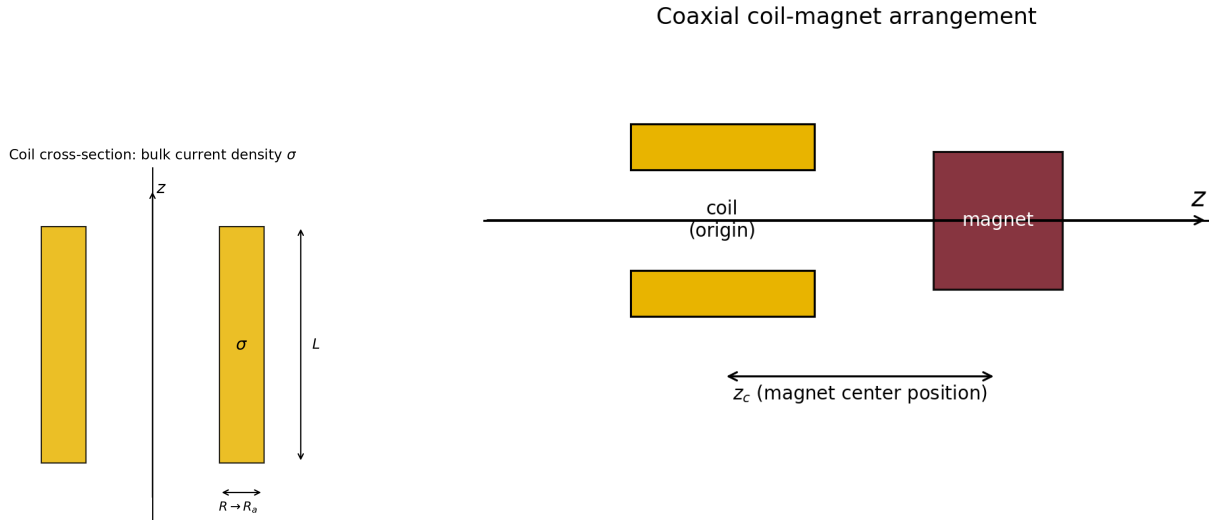


Figure 1: Left: coil cross-section carrying bulk azimuthal current density σ . Right: coaxial coil–magnet arrangement; the magnet center position z_c is the swept variable.

Coil. Rather than resolving N discrete turns, the coil is smeared into a bulk azimuthal current density

$$\sigma = \frac{NI}{V_{\text{coil}}}, \quad V_{\text{coil}} = \pi (R_a^2 - R^2) L, \quad (1)$$

i.e. σ is an *ampere-turn density* (A/m^3): integrating it over the coil’s volume returns the total ampere-turns NI .

Magnet. The magnet is treated as a continuum of axial point dipoles with dipole-moment density equal to its magnetization M_z (A/m). Each volume element dV at (r_2, z_2) carries moment $M_z dV$ and feels a force from the local gradient of the coil’s field.

2 Field of the Coil

2.1 One filamentary ring

Consider a single current loop of radius r_1 at height z_1 , carrying current I . Parametrize the loop by the azimuthal angle θ *relative to the field point* (Fig. 2) and apply Biot–Savart’s law directly in cylindrical coordinates — without reducing to elliptic integrals K, E . The separation between a source filament and a field point (r_2, z_2) is

$$D^2 = r_1^2 + r_2^2 - 2r_1r_2 \cos \theta + (z_2 - z_1)^2, \quad (2)$$

and the axial field the loop produces at that point is the standard raw Biot–Savart result

$$B_z(r_2, z_2) = \frac{\mu_0 I r_1}{4\pi} \int_0^{2\pi} \frac{r_1 - r_2 \cos \theta}{D^3} d\theta. \quad (3)$$

Filament pair separation $D(r_1, r_2, \theta, \Delta z)$

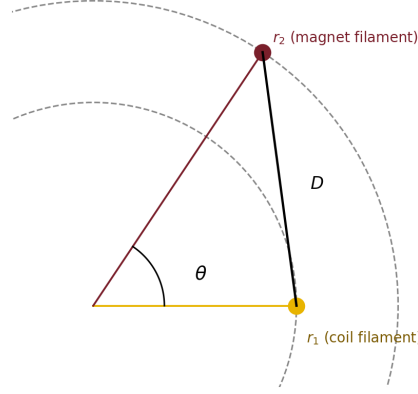


Figure 2: Relative azimuth θ between a coil filament (radius r_1) and a magnet filament (radius r_2); their separation is D .

2.2 Smearing over the coil cross-section

A thin ring of the coil occupying $[r_1, r_1 + dr_1] \times [z_1, z_1 + dz_1]$ sweeps out a volume $2\pi r_1 dr_1 dz_1$, so by Eq. (1) it carries current

$$dI = \sigma (2\pi r_1) dr_1 dz_1. \quad (4)$$

Substituting Eq. (4) into Eq. (3) and integrating over the coil’s cross-section gives the field of the *whole* coil at an arbitrary off-axis point:

$$B_z(r_2, z_2) = \frac{\mu_0 \sigma}{2} \int_R^{R_a} \int_{-L/2}^{L/2} r_1^2 \int_0^{2\pi} \frac{r_1 - r_2 \cos \theta}{D^3} d\theta dz_1 dr_1. \quad (5)$$

3 Force on the Magnet

3.1 Field gradient

The force on a dipole is $dF_z = M_z (\partial B_z / \partial z_2) dV$. Differentiating Eq. (2)–(5) with respect to z_2 (only D depends on it) brings down one power of $(z_2 - z_1)$ and raises the power of D in the denominator by two:

$$\frac{\partial}{\partial z_2} (D^{-3}) = -3(z_2 - z_1) D^{-5}, \quad (6)$$

so that

$$\frac{\partial B_z}{\partial z_2} = -\frac{3\mu_0\sigma}{2} \int_R^{R_a} \int_{-L/2}^{L/2} r_1^2 \int_0^{2\pi} \frac{(r_1 - r_2 \cos \theta)(z_2 - z_1)}{D^5} d\theta dz_1 dr_1. \quad (7)$$

3.2 Integrating over the magnet volume

Integrate the force density $M_z \partial B_z / \partial z_2$ over the magnet, $dV = r_2 dr_2 dz_2 d\phi_2$. The integrand does not depend on the magnet's own azimuth ϕ_2 , so that integral is trivial and contributes a factor 2π :

$$F_z(z_c) = 2\pi M_z \int_0^a \int_{z_c-h/2}^{z_c+h/2} r_2 \frac{\partial B_z}{\partial z_2} dz_2 dr_2. \quad (8)$$

4 Result

Substituting Eq. (7) into Eq. (8) collapses the 2π and the $3/2$ into a clean prefactor, giving the full five-fold integral:

$$F_z(z_c) = -3\pi\mu_0\sigma M_z \int_{z_c-h/2}^{z_c+h/2} \int_0^a \int_{-L/2}^{L/2} \int_R^{R_a} \int_0^{2\pi} \frac{r_1^2 r_2 (r_1 - r_2 \cos \theta) (z_2 - z_1)}{D^5} d\theta dr_1 dz_1 dr_2 dz_2 \quad (9)$$

with D given by Eq. (2). Structurally, the outer pair of integrals (z_2, r_2) ranges over the magnet's cross-section; the inner pair (z_1, r_1) ranges over the coil's cross-section; and the innermost integral (over θ) accounts for the relative azimuth between a coil filament and a magnet filament.

5 Numerical Evaluation

No closed form (elliptic-integral or otherwise) is substituted anywhere — Eq. (9) is evaluated exactly as written, by nested Gauss–Legendre quadrature. Each of the five integration variables is given its own set of quadrature nodes and weights, mapped from the reference interval $[-1, 1]$ onto its physical range:

Variable	Range
θ	$[0, 2\pi]$
r_1	$[R, R_a]$
z_1	$[-L/2, L/2]$
r_2	$[0, a]$
z_2	$[z_c - h/2, z_c + h/2]$

Sweeping z_c over the scan range and repeating this quadrature at each point produces the F_z vs. z_c curve, whose extremum identifies the force-maximizing (“sweet spot”) magnet position.

Note. This is the *exact* extended-body force — no assumption is made that the magnet is small compared to the coil's field variation. A point-dipole model (evaluating $\partial B_z / \partial z_2$ at a single point, the magnet's center, and multiplying by its total moment $M_z V_{\text{magnet}}$) is a common simplification but is only accurate in the limit where the magnet is much smaller than the length scale over which the coil's field changes — a limit not always satisfied in a compact BOSEM geometry.